Kinematics Problems

1. A ball is thrown vertically upward from a 2nd story window with an initial velocity of 10 m/s. If a person in a third story window that is 5.0 m above the first window catches the ball on its way down, how long was the ball in the air? [1.2 s]

2. If the sum of two vectors equals zero, the magnitude of their difference is
   a) equal to the magnitude of either one
   b) equal to twice the magnitude of either one
   c) less than twice the magnitude of either one.

3. A turtle is moving with constant acceleration along a straight ditch. He starts his stopwatch as he passes a fence post and notes that it takes him 10s to reach a pine tree 10m farther along the ditch. As he passes the pine tree his speed is 1.2 m/s. How far was he from the fence post when he started from rest? (SIN 76) [ans: 8.0 m]

4. A British Concorde (BC) and a French Concorde (FC) flew in opposite directions around the earth (40 000 km). The BC covered half of its flight distance at a supersonic speed of 2500 km/h and the other half at a subsonic speed of 1000 km/h. The FC spent half of its flight time at 2500 km/h and the other half at 1000 km/h. Which Concorde completed the trip first, and by how many hours did it beat the other? [FC by 5.1 h]

5. The Easter bunny runs along a straight and narrow path with a constant speed of 25m/s. He passes a sleeping tortoise, which immediately starts to chase the bunny with a constant acceleration of 3.0×10^{-3} m/s^2. How long does it take the tortoise to catch the bunny? [4.6 h]

6. A person travelling between London and Windsor (200 km apart) wants to complete a round trip at an average speed of 100 km/h. On arriving at Windsor at the end of the first leg, she discovers that she has only averaged 75km/h. What must her average speed by on the return trip? (CAP Contest 1984) [150 km/h]

7. A car going 80.0 km/h overtakes a 1.20 km long train that is travelling at 65 km/h in the same direction on a track parallel to the road. How long does it take the car to pass the train? How long would it take if the vehicles were approaching each other travelling in opposite directions? [4.8 min; 0.50 min]

work on following pages

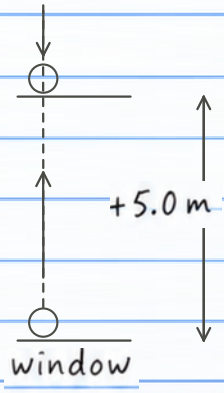
Problem Solving

1) + is up

$v_i = 10\ m/s$

$a = -9.81\ m/s^2$

$\Delta d = 5.0\ m$



$$\Delta d = v_i \Delta t + \frac{1}{2} a \Delta t^2$$

$$5.0 = 10\Delta t - 4.905\Delta t^2$$

$$4.905\Delta t^2 - 10\Delta t + 5.0 = 0$$

use quadratic formula

$$\Delta t = \frac{10 \pm \sqrt{100 - 4(4.905)(5.0)}}{9.81}$$

$\Delta t = 0.88\ s$ or $1.16\ s$

use the later time since it is caught going down

$\Delta t = 1.2\ s$

2) $\vec{A} + \vec{B} = 0$, so $\vec{B} = -\vec{A}$

$\vec{A}$ $\vec{B}$

(equal length)

$$\vec{A} - \vec{B} = \vec{A} + (-\vec{B}) = 2\vec{A}$$

$\vec{A}$ $-\vec{B}$

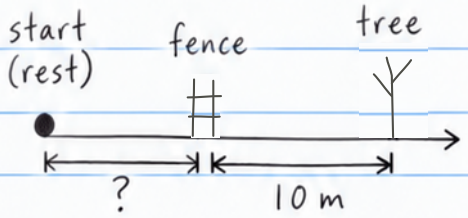
$2\vec{A}$

(b) twice the magnitude

3)

$$\Delta d = \frac{1}{2}(v_{fence} + v_{tree})\,\Delta t$$

$$10 = \frac{1}{2}(v_{fence} + 1.2)(10)$$



$$v_{fence} = 0.80 \text{ m/s}$$

$$a = \frac{1.2 - 0.80}{10} = 0.040 \text{ m/s}^2$$

start to fence:

$$v_{fence}^2 = v_{start}^2 + 2a\,\Delta d$$

$$\Delta d = \frac{0.80^2 - 0^2}{2(0.040)}$$

$\Delta d = 8.0$ m before the fence

4)

British: half the distance at each speed

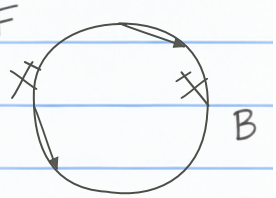
$$\Delta t_B = \frac{20000}{2500} + \frac{20000}{1000} = 28 \text{ h}$$

French: half the time at each speed

$$40000 = 2500\left(\frac{\Delta t_F}{2}\right) + 1000\left(\frac{\Delta t_F}{2}\right)$$

$$\Delta t_F = \frac{40000}{1750} = 22.86 \text{ h}$$

$$28 - 22.86 = 5.14 \text{ h}$$

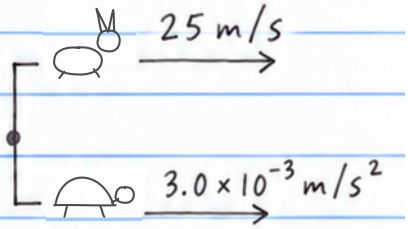


French Concorde, by 5.1 h

5) same position when they meet

$\Delta d_{bunny} = 25\Delta t$

$\Delta d_{tortoise} = \frac{1}{2}(3.0\times10^{-3})\Delta t^2$



$25\Delta t = 0.0015\Delta t^2$

$\Delta t(0.0015\Delta t - 25) = 0$

$\Delta t = 0$ or 1.67×10^4 s

0 is when they start, so use the later time

$\Delta t = 4.6$ h

6) find the total time first

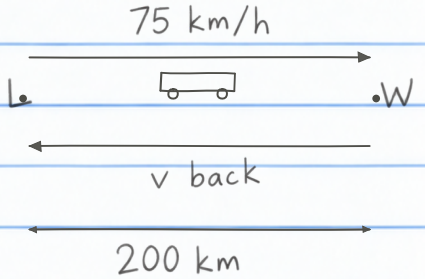
$\Delta t_{total} = \frac{(2\times200)}{100} = 4.0$ h

$\Delta t_{out} = \frac{200}{75} = 2.67$ h

$\Delta t_{back} = 4.0 - 2.67 = 1.33$ h

$V_{back} = \frac{200}{1.33}$

$V_{back} = 150$ km/h



7) + is the car's direction

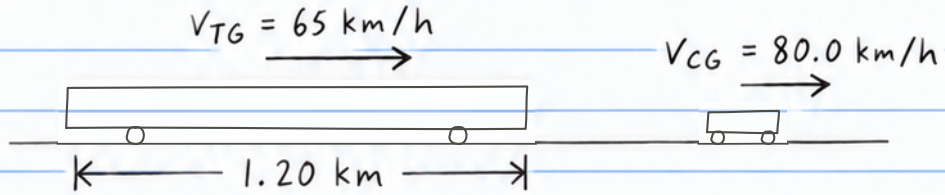
train length = 1.20 km

same direction:

$V_{CT} = V_{CG} - V_{TG} = 80.0 - 65 = 15 \text{ km/h}$

$\Delta t = \frac{L}{V_{CT}} = \frac{1.20}{15} = 0.080 \text{ h}$

$\Delta t = 4.8 \text{ min}$



opposite directions:

$V_{CT} = 80.0 - (-65) = 145 \text{ km/h}$

$\Delta t = \frac{1.20}{145} = 0.00828 \text{ h}$

$\Delta t = 0.50 \text{ min (about 30 s)}$

